

MA 102 Mathematics II
Ordinary Differential Equations
Lecture 4
(March 11, 2024)



Reduction of order:

The most general second-order differential equation has the form

$$F(x, y, y', y'') = 0.$$

Here, we consider two special types of second-order differential equations that can be solved by using the methods used for solving first-order differential equations.



I. Dependent variable absent:

If y is not explicitly present, the above equation can be written in the form

$$F(x, y', y'') = 0. \quad (1)$$

In this case, we introduce a new dependent variable p by putting

$$y' = p \quad \text{so that} \quad y'' = \frac{dp}{dx}. \quad (2)$$

This substitution transforms (1) into the following first-order equation:

$$F(x, p, \frac{dp}{dx}) = 0. \quad (3)$$

We can solve for p from the equation (3), and then by replacing p with dy/dx , we can solve for y .

This method reduces the problem of solving a second-order differential equation to solving two first-order differential equations in succession - one in p and then another in y .

Example:

Solve $xy'' - y' = 3x^2$.

Solution:

By putting

$$y' = p \quad \text{and} \quad y'' = \frac{dp}{dx},$$

The equation becomes

$$xp' - p = 3x^2.$$

The equation gives

$$\frac{dp}{dx} - \frac{1}{x}p = 3x.$$

The integrating factor can be obtained as

$$\text{I.F.} = e^{-\int \frac{1}{x} dx} = e^{-\ln x} = \frac{1}{x}.$$

The solution for p can be obtained as

$$\begin{aligned}p \times \frac{1}{x} &= 3 \int \frac{x}{x} dx + c \\&= 3x + c \\ \Rightarrow p &= 3x^2 + cx.\end{aligned}$$

Now since $p = y'$, we have another first-order equation:

$$\frac{dy}{dx} = 3x^2 + cx,$$

which can be solved to get

$$y = x^3 + \frac{1}{2}cx^2 + d.$$



II. Independent variable absent:

If x is not explicitly present, the most general second order equation can be written as

$$G(y, y', y'') = 0. \quad (4)$$

Consider

$$y' = p, \quad \text{and} \quad y'' = \frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dy}. \quad (5)$$

This enables us to write (4) in the form

$$G\left(y, p, p \frac{dp}{dy}\right) = 0, \quad (6)$$

and proceed to solve two first-order equations in succession - one in p and then another in y .

Example:

Solve $y'' + y = 0$.

Solution:

The equation can be written as

$$\begin{aligned} p \frac{dp}{dy} + y &= 0 \\ \Rightarrow p \, dp &= -y \, dy. \end{aligned}$$

Integrating

$$\begin{aligned} p^2 &= c^2 - y^2 \\ \Rightarrow p &= \sqrt{c^2 - y^2} \text{ taking the positive sign.} \end{aligned}$$

Since $p = dy/dx$:

we get

$$\frac{dy}{\sqrt{c^2 - y^2}} = dx.$$



Now integrating

$$\sin^{-1} \frac{y}{c} = x + d.$$

This gives us the solution as

$$y = c \sin(x + d).$$



By now, we realize that

there are four major types of first-order ordinary differential equations: **separable**, **homogeneous**, **exact** and **linear**.

There are some nonlinear differential equations which can be reduced to linear form by suitable change of the dependent variable.

We will discuss three special types of nonlinear first-order ordinary differential equations and their solution techniques:

- 1 Bernoulli's equation
- 2 Riccati equation
- 3 Equation of Clairaut



I. Bernoulli's equation

Bernoulli's equation has the following form:

$$y' + P(x)y = Q(x)y^n, \quad (7)$$

where n may take any value. If $n = 0$ or $n = 1$, this equation becomes linear.

For all other values of n , (7) is nonlinear. It can be written as

$$y^{-n}y' + P(x)y^{1-n} = Q(x). \quad (8)$$

Affecting a change of the dependent variable by $z = y^{1-n}$ so as to have

$$z' = (1 - n)y^{-n}y'.$$

Substituting in (8):

$$z' + (1 - n)P(x)z = (1 - n)Q(x), \quad (9)$$

which is a linear first-order equation in the new dependent variable z instead of the original dependent variable y . This equation can be solved by the method adopted earlier.

The solution to the original problem is given by

$$y = z^{1/(1-n)}. \quad (10)$$

Example

$$3xy' + y + x^2y^4 = 0.$$

Solution:

The given equation can be written in Bernoulli's form as

$$\begin{aligned} y' + \frac{1}{3x}y &= -\frac{x}{3}y^4 \\ \Rightarrow y^{-4}y' + \frac{1}{3x}y^{-3} &= -\frac{x}{3} \end{aligned}$$

Putting $z = y^{-3}$ so as to have

$$z' = -3y^{-4}y'.$$



Then the ODE gets transformed to

$$z' - \frac{1}{x}z = x.$$

The integrating factor is

$$\text{I.F.} = e^{-\int \frac{1}{x} dx} = e^{-\ln x} = \frac{1}{x}.$$

Therefore, the solution can be written as

$$\begin{aligned} z \times \frac{1}{x} &= \int \frac{x}{x} dx + c = x + c \\ \Rightarrow z &= x(c + x). \end{aligned}$$

Therefore, the required solution is

$$y^{-3} = x(c + x),$$

or,

$$y^3 = \frac{1}{x(c + x)}.$$



Riccati equation

Another first-order ordinary differential equation of importance is

$$y' = P(x)y^2 + Q(x)y + R(x), \quad (11)$$

where P , Q and R are functions of x . For $R(x) = 0$, equation (11) reduces to Bernoulli's equation.

For $R(x) \neq 0$, we develop the following method for finding solution to (11).

Let $y = u(x)$ be a known solution of (11). Then put

$$y = u + \frac{1}{z}, \quad (12)$$

where $z(x)$ is an unknown function.

Then

$$y' = u' - \frac{1}{z^2}z'.$$



Then, equation (11) becomes

$$u' - \frac{1}{z^2}z' = P(u^2 + \frac{1}{z^2} + \frac{2u}{z}) + Q(u + \frac{1}{z}) + R.$$

We obtain

$$u' - \frac{1}{z^2}z' = (Pu^2 + Qu + R) + P(\frac{1}{z^2} + \frac{2u}{z}) + \frac{Q}{z}. \quad (13)$$

Since u satisfies $u' = Pu^2 + Qu + R$, we have

$$-\frac{dz}{dx} = P(1 + 2uz) + Qz, \quad (14)$$

which is a linear differential equation in z since u is a known quantity.

(14) can be seen as

$$\frac{dz}{dx} + (2Pu + Q)z = P(x), \quad (15)$$

which can be solved by using the usual procedure and then get $y = u + \frac{1}{z}$ as the solution.

Example:

$$y' = y^2 + (1 - 2x)y + (x^2 - x + 1); \quad u = x.$$

Solution:

Here $P = 1$, $Q = 1 - 2x$ and $R = x^2 - x + 1$.

Let

$$y = x + \frac{1}{z}$$

so that

$$y' = 1 - \frac{1}{z^2} z'.$$

Therefore

$$\begin{aligned} 1 - \frac{1}{z^2} z' &= \left(x + \frac{1}{z}\right)^2 + (1 - 2x)\left(x + \frac{1}{z}\right) + (x^2 - x + 1) \\ \Rightarrow 1 - \frac{1}{z^2} z' &= x^2 + \frac{2x}{z} + \frac{1}{z^2} + x + \frac{1}{z} - 2x^2 - \frac{2x}{z} + x^2 - x + 1. \end{aligned}$$

This gives us

$$\begin{aligned}-\frac{1}{z^2}z' &= \frac{1}{z} + \frac{1}{z^2} \\ \Rightarrow z' &= -z - 1.\end{aligned}$$

Integrating with integrating factor e^x :

$$z = Ae^{-x} - 1.$$

The solution can be written as

$$\begin{aligned}y &= x + \frac{1}{z} \\ \Rightarrow y &= x + \frac{1}{Ae^{-x} - 1},\end{aligned}$$

where A is an arbitrary constant.



III. Equation of Clairaut

is as follows:

$$y = xy' + f(y'). \quad (16)$$

The solution of equation (16) can be written as

$$y = mx + f(m), \quad (17)$$

where m is a parameter.

Since $y' = m$, by eliminating m from (17), we get the ODE (16).

Example:

$$y = xy' + (y')^2.$$

Solution:

Let $y' = v$ so that $y = xv + v^2$.

Differentiating

$$y' = v + xv' + 2vv'.$$

Since $y' = v$, we have

$$v'(x + 2v) = 0.$$

There are two cases to consider:

Case I: $v' = 0 \Rightarrow v = m$. This gives the general solution

$$y = mx + m^2.$$

Case II:

$$x + 2v = 0 \Rightarrow v = -\frac{x}{2}.$$

Substituting in $y = xv + v^2$ yields

$$y = -\frac{x^2}{4}.$$

This solution satisfies the differential equation but cannot be obtained from $y = mx + m^2$ by giving particular values of m .

The solution $y = -x^2/4$ is usually called a **singular solution** which does not contain any arbitrary constant.

It can be analytically proved that $y = mx + m^2$ is a tangent line which touches the parabola $y = -x^2/4$. Thus, for different values of m , we get different tangent lines, and the parabola $y = -x^2/4$ which envelopes all tangent lines is called the **envelope**.